

GA2 - Tema seminar 1

$$1. (A+B)^h = C_h^0 A^h B^0 + C_h^1 A^{h-1} B^1 + \dots + C_h^{h-1} A^1 B^{h-1} + C_h^h A^0 B^h$$

$$\text{cum } A^0 = B^0 = \bar{1}^h \Rightarrow$$

$$(A+B)^h = C_h^0 A^h + C_h^1 A^{h-1} B^1 + \dots + C_h^{h-1} A^1 B^{h-1} + C_h^h B^h$$

$$\text{cum } C_h^i = C_h^{h-i}, \quad i \in \mathbb{N}, \quad i \neq 0, h \Rightarrow$$

$$\begin{aligned} (A+B)^h &= C_h^h A^h + C_h^{h-1} A^{h-1} B^1 + \dots + C_h^1 A^1 B^{h-1} + C_h^0 B^h \\ &= \sum_{j=0}^h C_h^j A^j B^{h-j} \end{aligned}$$

$$\begin{aligned} 2. \quad A+B &= B+A \\ A+B &= AB \\ B+A &= BA \end{aligned} \quad \parallel \Rightarrow \quad AB = BA$$

$$3. \quad A = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \Rightarrow A^n = \begin{pmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{pmatrix} \quad (\forall) n \in \mathbb{N}^*$$

Aplicăm inducția matematică:

$$P(n): A^n = \begin{pmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{pmatrix} \quad (\forall) n \in \mathbb{N}^*$$

Etapa de verificare

$$P(1): A^1 = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \quad (\text{din ipoteza } "A")$$

$$P(2): A^2 = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$$A^2 = \begin{pmatrix} \cos^2 x - \sin^2 x & 2 \sin x \cos x \\ -2 \sin x \cos x & \cos^2 x - \sin^2 x \end{pmatrix}$$

$$\text{cum } \begin{cases} \sin 2x = 2 \sin x \cos x \\ \cos 2x = \cos^2 x - \sin^2 x \end{cases} \Rightarrow$$

$$\Rightarrow A^2 = \begin{pmatrix} \cos 2x & \sin 2x \\ -\sin 2x & \cos 2x \end{pmatrix} = A^1$$

Etapa de inducție propriu-zisă:

presupunem $P(h)$, $h \in \mathbb{N}^+$ propoziție adevărată, dem.

că $P(h+1)$ prin A^h :

$$P(h+1): A^{h+1} = A^h \cdot A$$

$$A^{h+1} = \begin{pmatrix} \cos h x & \sin h x \\ -\sin h x & \cos h x \end{pmatrix} \begin{pmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{pmatrix}$$

$$= \begin{pmatrix} \cos h x \cos x - \sin h x \sin x & \cos h x \sin x + \sin h x \cos x \\ -(\cos h x \sin x + \sin h x \cos x) & \cos h x \cos x - \sin h x \sin x \end{pmatrix}$$

$$\text{cum } \begin{cases} \cos(a+b) = \cos a \cos b - \sin a \sin b \\ \sin(a+b) = \sin a \cos b + \cos a \sin b \end{cases} \Rightarrow$$

$$\Rightarrow A^{h+1} = \begin{pmatrix} \cos (h+1)x & \sin (h+1)x \\ -\sin (h+1)x & \cos (h+1)x \end{pmatrix} \quad "A^h" \Rightarrow \text{inducție completă} \Rightarrow$$

$$\Rightarrow A^n = \begin{pmatrix} \cos n x & \sin n x \\ -\sin n x & \cos n x \end{pmatrix}, \quad n \in \mathbb{N}^+$$

$$4. \quad A = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{vmatrix}$$

$$\det A = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{vmatrix} \begin{matrix} 3 \\ 4 \\ -1 \\ 4 \end{matrix} \begin{matrix} \underline{\underline{C_2 = C_2 - C_1}} \\ C_3 = C_3 - 2C_1 \\ C_4 = C_4 - 3C_1 \end{matrix} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 2 & 3 & -3 & -7 \\ -1 & -1 & 4 & 7 \end{vmatrix} =$$

$$\begin{aligned} &= 1 - (-1)^{1+1} \begin{vmatrix} 0 & 1 & 1 \\ 3 & -3 & -7 \\ -1 & 4 & 7 \end{vmatrix} = 0 + 7 + 12 - 3 - 21 - 0 = -5 \\ &\text{devol} \end{aligned}$$